

Randomly pivoted Cholesky and structured sketches: results, connections, and open questions

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Abstract

Six companion papers study the probability laws behind low-rank approximation from restricted matrix information. Their principal results are an all-budget comparison between Haar-frame pivoting and volume sampling, an order-one drift bound for arbitrary positive-semidefinite matrices under product probes, and robustness bounds for approximate and sketched pivot probabilities. Further papers characterize exact Gaussian range laws and refine SparseStack and two-round SRHT prescriptions. This overview states the results with their hypotheses, identifies the established tools they use, and explains the remaining gaps. In particular, the $O(r/\varepsilon)$ guarantee for original coordinate pivoting remains open.

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1 The question and its established benchmarks

For a positive-semidefinite matrix A over $\mathbb{R}$ or $\mathbb{C}$, put $\tau_r(A) = \sum_{j>r} \lambda_j(A)$, where the eigenvalues are decreasing. This is the best rank- r trace error. Original randomly pivoted Cholesky starts at $R_0 = A$, samples a coordinate with probability $\pi_i = (R_t)_{ii} / \text{tr } R_t$, and updates

$$R_{t+1} = R_t - \frac{R_t e_i e_i^* R_t}{(R_t)_{ii}}. \quad (1)$$

A zero residual is absorbing. Chen, Epperly, Tropp, and Webber developed the practical algorithm and its trace analysis [1]. Its basic progress identity is

$$\mathbb{E}[\text{tr } R_{t+1} \mid R_t] = \text{tr } R_t - \frac{\text{tr}(R_t^2)}{\text{tr } R_t}. \quad (2)$$

This identity supplies an efficient completion once the trace is comparable to the target tail. By itself it retains dependence on the initial ratio $\text{tr } A / \tau_r(A)$.

Conjecture 1.1. *There is an absolute constant C such that every finite PSD input satisfies*

$$\mathbb{E} \text{tr } R_k \leq (1 + \varepsilon) \tau_r(A) \quad \text{for } k \geq Cr/\varepsilon, \quad 0 < \varepsilon \leq 1.$$

The process terminates exactly if the budget reaches rank A .

Epperly's recent analysis supplies a spectrum-independent pivot bound [2]. The supporting manuscript [13] claims the sharper bound

$$K_4(r, \varepsilon) = O(r/\varepsilon + r\Lambda_4(r)), \quad \Lambda_1(r) = \log(e + r), \quad \Lambda_{j+1}(r) = \log(e + \Lambda_j(r)).$$

The robustness companion imports its specified moment and finite-count statements. This overview summarizes that dependency and does not provide an independent proof of the supporting manuscript.

Volume sampling provides a different law and a useful benchmark. With $e_k(A)$ the k th elementary symmetric polynomial of the eigenvalues, its expected residual trace obeys the established identity and tail bound [3, 5]

$$V_k(A) = (k+1) \frac{e_{k+1}(A)}{e_k(A)} \leq \frac{k+1}{k+1-r} \tau_r(A), \quad k \geq r. \quad (3)$$

Set $V_k(A) = 0$ after exact recovery. Thus volume sampling reaches the desired error at $k = r - 1 + \lceil r/\varepsilon \rceil$. The factorial likelihood comparison between adaptive sampling and volume sampling is also established [4]; applying it in a new setting does not make the inequality itself a new result.

2 Random frames: a comparison at every budget

Let Q be a Haar orthogonal or Haar unitary matrix, independently of the pivots. The random-frame paper [16] proves

$$\mathbb{E}_{Q, \text{RP}} \text{tr } R_k(QAQ^*) \leq V_k(A) \quad \text{for every } k \geq 0. \quad (4)$$

Together with (3), this gives the volume-sampling pivot budget in expectation. The averaging covers both the frame and the sequential pivot process.

The proof reduces the residual spectrum to a Markov chain with Gamma coordinates. A covariance inequality for ratios of those coordinates then compares a transition with the elementary-symmetric benchmark. Existing volume-sampling identities concern determinant-weighted subsets; the contribution here is the comparison for Haar-averaged sequential pivoting. The paper also studies a Dirichlet family containing the real and complex cases. Monotonicity within that family remains a conjecture.

The limitation is computational. Dense Haar rotation can destroy cheap column access. Consequently (4) does not settle Conjecture 1.1, and it does not prove the same comparison for a Hadamard, sparse, or tensor-product frame. Coordinate examples with a 2^r expected-error factor at the exact budget r , due to Colbrook [6], show that the frame can matter strongly. The average-frame comparison shows that every fixed spectrum nevertheless meets the volume benchmark after averaging over frames.

3 Product probes: progress without an embedding hypothesis

On the tensor space $\bigotimes_{j=1}^q \mathbb{R}^{n_j}$, a legal probe has the form $z = x_1 \otimes \cdots \otimes x_q$. The product-probe paper [17] studies independent uniform spherical factors. Gaussian factors give the same Nyström output because nonzero column scales leave the selected range unchanged.

The main inequality holds for every PSD residual:

$$\mathbb{E}(\text{tr } R - \text{tr } R') \geq \frac{\text{tr}(R^2)}{\kappa \text{tr } R}, \quad \kappa = \exp \left(\sum_{j=1}^q D_{n_j} \right). \quad (5)$$

For the real and complex spherical laws, respectively,

$$\begin{aligned} D_n^{\mathbb{R}} &= \log n + \psi(3/2) - \psi(n/2 + 1), \\ D_n^{\mathbb{C}} &= \log n + 1 - H_n. \end{aligned}$$

Here ψ is the digamma function and H_n is the harmonic number. The complex branch on a real oracle carries the simulation cost stated in the companion; it is not counted as one free real product query.

The proof changes measure by the progress numerator and takes a fractional moment limit at order one. Summing the progress gives the whole-input bound

$$\mathbb{E} \|A - \hat{A}_m\|_F^2 \leq \frac{\kappa}{m} (\text{tr } A)^2. \quad (6)$$

For $\tau_r(A) > 0$, scalar completion gives a sufficient budget

$$m \geq \left\lceil \kappa r \left[\varepsilon^{-1} + \log \varepsilon^{-1} + \log \frac{\text{tr } A}{\tau_r(A)} \right] \right\rceil$$

for expected trace error $(1 + \varepsilon)\tau_r(A)$. The initial trace-ratio logarithm is therefore still present.

The product-query model and the limitations of Hutchinson-type averaging precede this work [7, 8]. Holden’s probability-sensitive probes and diagonal correction are particularly close prior results [9]. The companion distinguishes those ingredients from its progress-based

Nyström reduction and proves its displayed budget comparisons. The constant κ remains exponential in tensor order; its sharp drift exponent is only bracketed. This is not a minimax characterization of unrestricted adaptive product-query trace estimation.

4 Exact Gaussian range laws: a structural boundary

Let V be an isometry onto the full relevant support, of dimension d . For independent copies of a structured probe ω , the range-law paper [18] characterizes equality with the dense-Gaussian range law by

$$V^\top \omega \neq 0 \text{ almost surely,} \quad [V^\top \omega] \text{ uniform on } \mathbb{RP}^{d-1}. \quad (7)$$

Brackets identify vectors differing by a nonzero scalar. Independence between sketch columns and a condition on the complete support are both essential. A condition only on the leading eigenspace does not identify the full-input output law.

Uniform projected lines produce Gaussian range laws at every sketch size; conversely, the size-one law recovers the condition. Random column magnitudes are invisible to range-determined outputs, including PSD Nyström approximation. They need not be invisible to embedding estimates or to an uncalibrated two-sided formula.

Conditional scalar Gaussianity is a checkable sufficient mechanism: after conditioning on the other factors, the projected probe is Gaussian with covariance wI_d , where $w > 0$. The companion derives exact algorithm laws from this mechanism. The Gaussian pseudoinverse and inverse-Wishart formulas used afterward are standard [10]; the structured criterion and its consequences are the contribution.

The class is restrictive. Product probes satisfy the projective dimension ceiling

$$d \leq 1 + \sum_{j=1}^q (n_j - 1).$$

For the conditionally Gaussian class treated in the paper, a frozen product environment further reduces supported matrix-vector products to an n_1 -dimensional problem, and n_1 suitable product queries recover the input exactly. No growing-rank theorem for general tensor network sketches is claimed. A complex extension requires directly verified Hermitian scalar covariance; a real symmetric covariance identity alone is insufficient.

5 SparseStack and SRHT: parameters and whole-sketch energy

The two sketch companions [19, 20] use operator estimates already present in the repository [14, 15]. Their contributions refine scalar calibration and applications of those estimates. The base constructions and their order-optimal embedding rates are prior results.

5.1 An exact SparseStack constant certificate

SparseStack has s normalized CountSketch blocks, b buckets per block, and $m = sb$ actual rows. Put

$$q = \max\{1, \lceil \log_4(d/\delta) \rceil\}, \quad v = 2q + 1, \quad D = d + v.$$

The companion certifies the choices

$$(C, B) \in \{(69, 8), (74, 15/2), (79, 7)\}, \quad s = \lceil Bv/\varepsilon \rceil, \quad b = \left\lceil \frac{CD}{s\varepsilon^2} \right\rceil,$$

with row accounting

$$m < \frac{Cd + (2C + 2B)q + C + B + 1}{\varepsilon^2}.$$

An exact rational interval certificate supplies the sharper scalar step. The imported operator majorant is identified in the graded framework, rather than in an unavailable newer private draft.

For $Z = \Pi^\top \Pi - I$ and PSD A , a separate exact calculation gives

$$\mathbb{E} \operatorname{tr}(AZAZ) = \frac{(\operatorname{tr} A)^2 + \|A\|_F^2 - 2 \sum_i A_{ii}^2}{m}. \quad (8)$$

A deterministic inequality bounds the squared Frobenius Nyström residual by this energy. The result counts all actual rows and is rank-blind: it does not assert an expected error relative to a rank- r tail.

5.2 A canonical two-round SRHT prescription

For the two-round flat-transform model, the exact row-factor allowance Λ_q obeys $D_q \leq \Lambda_q \leq 3D_q$, where $D_q = d + 12q^2$ in the Walsh case. The companion's canonical prescription is

$$q_* = \max \left\{ 1, \left\lceil \log_4 \frac{6d}{5\delta} \right\rceil \right\},$$

$$M = \min \left\{ n, \left\lceil \frac{20\Lambda_{q_*}}{\varepsilon^2} \right\rceil \right\}.$$

The sign independence level is $\min(n, 4q_*)$. At $M = n$ the full unitary transform is retained, so the embedding is exact. The simpler coherence prescription is $60D_q/\varepsilon^2$.

The exact Bernoulli-to-fixed-size barycenter is already an ingredient of the base analyses [14, 15]. The refinement retains its sharper factor during scalar calibration. Yang independently analyzes the rerandomized two-Walsh construction with a linear-in-dimension embedding prescription [12]; the companion credits that result and isolates its own explicit-confidence and constant refinements.

For a uniformly subsampled real flat transform with the stated sign independence, the companion also proves

$$\mathbb{E} \|K - \operatorname{Nys}_K(S^*)\|_F^2 \leq \frac{2}{M} \left(1 - \frac{M}{n}\right) (\operatorname{tr} K)^2, \quad K \succeq 0.$$

This is an absolute energy bound. Its finite-population factor makes it vanish when all rows are retained. Nonadaptive Nyström trace estimation is itself established [11]; the new application here uses the structured shared-row energy calculation.

5.3 A fixed-reference gate for adaptive frame selection

Both companions analyze a fixed range and a fixed dictionary of frames. The selecting RPCholesky law can depend on the sketch through a positive-definite coefficient matrix C . A sketch-independent volume reference dominates that law with likelihood price

$$\Lambda_C = \frac{k! \det C}{\prod_{j=0}^{k-1} \tau_j(C)} \leq k!.$$

This permits the specified sketch-dependent selection without a union bound over all frames. The likelihood inequality is the established Deshpande–Vempala comparison [4]; the fixed-reference reuse application and its parameter consequences are the contributions here. The theorem does not cover arbitrary learned subspaces or arbitrary adaptive sketch reuse.

6 Approximate and sketched pivot probabilities

For $0 < a \leq 1$, the robustness companion [21] separates three conditions on the actual pivot probabilities:

Condition	Proved consequence in the companion
$a\pi_i \leq q_i \leq \pi_i/a$	The imported four-logarithm conversion survives with explicit losses in a .
$q_i \geq a\pi_i$	Scalar trace completion survives with effective rank r/a ; the initial trace-ratio logarithm remains.
$q_i \leq \pi_i/a$	For $a < 1$, diagonal inputs prevent a uniform accuracy factor below $1/a$ at any fixed finite pivot budget.

The third conclusion concerns worst-case inputs at a fixed budget. It does not say that a particular finite matrix cannot eventually be recovered exactly. The two-sided pivot-count theorem already appears as Proposition 8.1 of the supporting paper [13]; the companion recalls it and isolates its functional hypotheses. Its further contributions include the one-sided distinction and the reused-sketch transfer below.

For sketched pivoting, write $A = F^*F$ and run the pivot process on $(\Pi F)^*(\Pi F)$. The proof uses embeddings of fixed low-dimensional spaces along a reference path. A hybrid process switches to exact pivoting at the first failed embedding event. Its likelihood comparison yields

$$\mathbb{E} \operatorname{tr} R_A(J) \leq (1 + \varepsilon) \tau_r(A) + P_{\text{fail}} \operatorname{tr} A. \quad (9)$$

The additive term is part of the guarantee. An expectation relative to the tail requires enough confidence to make it small relative to $\tau_r(A)$.

At fixed sketch distortion η , the hybrid conversion retains the four-logarithm pivot dependence. Under the stated choices, the SparseStack row budget has order

$$\frac{K + \log(\operatorname{tr} A / (\varepsilon \tau_r(A)))}{\eta^2}, \quad \tau_r(A) > 0.$$

The SRHT confidence allowance contains its quadratic logarithmic term; a general linear-in- K claim for both sketches would be incorrect. The paper’s union-of-independent-chains inequality

is proved, but the $O(r/\varepsilon)$ restart consequence still requires a uniform fractional warm-start conjecture.

7 Connections and open questions

The connections in Table 1 distinguish established ingredients from the contributions being proposed.

Companion	Established ingredient	Contribution
Random frames	Volume expectation and tail inequality	All-budget comparison for Haar-averaged sequential pivoting.
Product drift	Product-query model, directional moments, diagonal correction	Order-one progress and Nyström consequences.
Range laws	Gaussian range invariance and pseudoinverse moments	Structured projective criterion and exact-law reductions.
SparseStack	CountSketch and graded majorants	Exact scalar certificate and whole-sketch energy consequences.
Two-round SRHT	Fast transforms, moment kernels, exact barycenter	Sharper calibration and explicit prescriptions.
Robust pivots	Coordinate drift, four-logarithm moments, and the existing two-sided bracket theorem	Functional conversion, one-sided distinction, and path-local sketch transfer.

Table 1: Mathematical ingredients and the isolated contributions.

The following questions remain separate mathematical targets:

1. Does original coordinate RPCholesky satisfy Conjecture 1.1?
2. Can an inexpensive structured frame preserve the all-budget volume comparison, exactly or with a constant loss?
3. What is the sharp product-probe drift exponent, and what is the unrestricted adaptive product-query complexity of trace estimation?
4. How stable is the exact Gaussian range law under approximately scalar conditional covariance?
5. Can path-local sketch failure prices be reduced enough to explain empirical sketch sizes while controlling (9)?

The numerical illustrations in the companions do not settle these questions. The supporting four-logarithm paper and the new companions are informal manuscripts; their use here does not assert formal verification or external peer review.

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